MILITARY SPECIFICATION

ENGINE INSTRUMENTATION SYSTEM,
AIRCRAFT A/A245-11

This amendment forms a part of MIL-E-27669B(USAF) and is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 2.1 Delete "first paragraph" and add: "Government documents".

* Add: "2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation."

* Under SPECIFICATION, Federal, delete "QQ-P-416" and add: "PPP-B-601 Boxes, Wood, Cleated Plywood"; "PPP-B-636 Box, Shipping, Fiberboard".

* Under SPECIFICATION, Military, delete "MIL-E-5272" and add: "DOD-D-1000 Drawings, Engineering and Associated Lists".

* Add the following boxed paragraph to the bottom center of page:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

PAGE 2

* Under SPECIFICATIONS, Military, delete "MIL-P-7936", "MIL-M-26512" and "MIL-D-70327". Add the following "MIL-M-7793 Meter, Time Totalizing", "MIL-B-27497 Bearing, Jewel, Sapphire Or Ruby, Synthetic" and "MIL-C-83488 Coating, Aluminum, ION Vapor Deposited".

* Under STANDARDS, Military, delete "MIL-STD-831", "MS17322", "MS33586" and add: "MIL-STD-129 Marking For Shipment And Storage" "MIL-STD-471 Maintainability Demonstration", "MIL-STD-810C Environmental Test Methods and "MIL-STD-889 Dissimilar Metals".

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:37Z
Check the source to verify that this is the current version before use.

Correct title as follows: "MIL-STD-701 List Of Standard Semiconductor Devices".

* Add: "2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)"

"2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained."

PAGE 3

* 3.1 Delete existing paragraph and add: "3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2 and 6.4)"

PAGE 4

* Add: "3.4.9 Recycled and reclaimed materials. Recycled and reclaimed materials shall be encouraged to the maximum extent possible."

PAGE 14

Add new paragraph as follows: "3.6.3 Synthetic jewel bearings. When synthetic sapphire or ruby bearings are used, they shall be in accordance with MIL-B-27497".

* 3.7.C Delete "Salt spray" and add "Salt fog". Delete "MIL-E-5272" and add: "MIL-STD-810C, Method 509.1"

PAGE 17

* 3.7.9a Delete "MIL-E-5272" and add: "MIL-STD-810C, Method 508.1".

* 3.7.9b Delete and add: "b. Dust (fine sand) in accordance with MIL-STD-810C, Method 510.1, procedure I".

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:37Z
Check the source to verify that this is the current version before use.

* 3.7.9c Delete "MIL-E-5272, procedure V" and add: "MIL-STD-810C, Method 516.2 procedure I".

* 3.7.9d Delete "MIL-E-5272" and add: "MIL-STD-810C, Method 507.1".

* 3.7.9e Delete "MIL-E-5272, procedure XII" and add: "MIL-STD-810C, procedure I, curve D."

PAGE 18

* 3.7.11a Delete entire paragraph and add: "a. Vibration. Vibration in accordance with MIL-STD-810C, Method 514.2, procedure I, part 2 only, curve B."

PAGE 19

3.8.2 Delete existing paragraph and add: "3.8.2 Converter channel design. Each converter channel shall be designed to drive two of its respective indicator channels through dual output receptacles wired in accordance with figures 4, 5, 6, and 7. System performance requirements specified herein shall be met when the converter channels are thus connected".

PAGE 24

* 3.8.2.7 Delete "MS17322-6" and add: "MIL-M-7793/6".

3.9.2 Line 7 delete "and filling tubes" Delete "sixth sentence from the paragraph".

PAGE 26

3.9.2.1 Delete existing paragraph and add: "3.9.2.1 Filling medium. The case shall be filled 100 percent nitrogen, This dry nitrogen filling gas must be at least 99.99 percent pure, free of dust particles, contain not more than 0.002 percent free oxygen and have a dew point not higher than -59.4°C (-75°F). The absolute pressure of the filling medium in the case shall be approximately one atmosphere".

3.9.2.2 Delete paragraph in its entirety.

PAGE 29

3.10 Line 4 delete reference to "MIL-D-70327" and add: "DoD-D-1000".

3.13 Delete existing paragraph and add: "3.13 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, where practical, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the preparing activity. Cadmium plating must be avoided when satisfactory alternative processes can be used".

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:37Z
Check the source to verify that this is the current version before use.

PAGE 31

* Add: "4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract.

Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material."

* 4.2(a)(b) Delete in its entirety and add: new "4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

* 4.3 Delete entirely and add new: "4.3 Inspection conditions. Unless otherwise specified all inspection shall be performed in accordance with the test conditions specified in this specification".

4.3.5 Line 6 delete "indicator" and add: "converter".

4.3.6.1 Line 5 delete "date" and add: "data".

PAGE 33

4.4 Change title of paragraph to "First article inspection".

* 4.4.2 Delete entire paragraph.

* 4.4.3 Title delete "Qualification tests." and add: "First article inspection".

4.4.3.1 Title delete "Qualification test schedule". and add: "First article inspection schedule".

* 4.5 Delete "acceptance tests" and add: "Quality conformance inspection" in two (2) places.

* 4.5 a. Delete "tests" and add: "inspection".

* 4.5 b. Delete "tests" and add: "inspections".

PAGE 40

* 4.6.1.4 Line 4 delete "while still at this temperature" and add: "while the converter is still at this temperature".

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:37Z
Check the source to verify that this is the current version before use.

* 4.6.1.5 Delete first sentence and add: "The converter shall be subjected to the vibration test specified in MIL-STD-810C, Method 514.2 procedure I, parts 1 and 2, curve B. Electronic assemblies shall be tested per part 2".

* 4.6.1.6 Delete "procedure V of MIL-E-5272" and add: "MIL-STD-810C, Method 516.2, procedure I".

PAGE 41

* 4.6.1.8 Delete "MIL-E-5272" and add: "MIL-STD-810C, Method 507.1, except at 95% humidity".

* 4.6.1.9 Delete "MIL-E-5272" and add: "MIL-STD-810C, Method 508.1".

* 4.6.1.10 Delete "spray" (2 places) and add: "fog". Delete "MIL-E-5272" and add: "MIL-STD-810C, Method 509.1".

* 4.6.1.11 Delete "Sand and dust" and add: "Dust (fine sand)". Delete "sand and dust" and add: "dust (fine sand)". Delete "MIL-E-5272" and add: "MIL-STD-810C, Method 510.1".

PAGE 42

4.6.2.4 Line 3 delete "$\pm 2^{o}$I" and add: "$\pm 2^{o}$C".

PAGE 43

4.6.2.8 Delete first sentence and add: "The indicators shall be subjected to a vibration test per MIL-STD-810C, Method 514.2, procedure I, part 2 only, curve B".

PAGE 44

4.6.2.11 Line 4 delete "0.1" and add: "1.0". Line 7, delete "1.0" and add: "10.0".

PAGE 45

4.6.2.13 Delete "spray" (2 places) and add: "fog". Delete "MIL-E-5272" and add: MIL-STD-810C, Method 509.1"

PAGE 51

4.6.3.14 Line 2 delete "MIL-M-26512" and add: "MIL-STD-471". Line 3, delete "Test and Demonstration" and add: "Selection".

* Delete Section 5 in its entirety and add:

5. Packaging

5.1 Preservation. Preservation shall be level A, C, or Industrial as specified. (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:37Z
Check the source to verify that this is the current version before use.

5.1.1 Level A.

5.1.1.1 Cleaning. Components of the Engine Instrumentation System shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Components of the Engine Instrumentation System shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each component of the instrumentaion system shall be packaged in quantity unit packs of one each in accordance with Method IC-I of MIL-P-116. Overbox completed pack in PPP-B-636 container. Apply sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the components of the engine instrumentation system when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each component of the engine instrumentation system shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of the component of the engine instrumentation system shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Components of the Engine Instrumentation System packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistant with the protection required.

5.2.2 Level B. Components of the engine instrumentaion system packaged as specified in 5.1.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged components of the engine instrumentation system shall be packed in accordance with ASTM D3951.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:37Z
Check the source to verify that this is the current version before use.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2) interior and exterior containers shall be marked in accordance with MIL-STD-129.

Add the following paragraphs:

"6.3.2 Converter channel. A converter channel is a single, self-sufficient, electrically isolated unit (entirely contained within the ECK-25 converter assembly) that receives an input signal and provides a processed signal through two outputs. One output drives the pilot's indicator channel and the other output drives the flight engineer's indicator channel.

6.3.3 Indicator channel. An indicator channel is one complete vertical scale display mechanism of a quadruple vertical scale tape indicator. Each indicator channel is electrically and mechanically isolated from the other three indicator channels contained within the indicator case.

6.3.4 System channel. A complete system channel for each parameter being measured is composed of the following:

a. A signal input for the parameter being measured.

b. A single converter channel with two outputs as defined in 6.3.2.

c. Two indicator channels as defined in 6.3.3 (one indicator channel is connected to the converter channel output for the pilot's usage, and the other indicator channel is connected to the converter channel output for the flight engineer's usage)".

6.4 Delete existing paragraph and add: "6.4 First article. First article test and inspection are required".

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodian:
AIR FORCE -99

Preparing activity:
AIR FORCE -71

Project number:
6620-F461

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:37Z
Check the source to verify that this is the current version before use.